

PD-PRUNE: Disaggregated Prefill and Decode Pruning for Efficient Agentic Inference

Anonymous ACL submission

Abstract

Large language models perform agentic inference by expanding a committed prefix into multiple candidate residuals and retaining a single trajectory. Existing search and early-stopping methods assign pruning scores only after tokens have been generated, so unpromising residuals still incur GPU prefill, KV transfer on a disaggregated connector, and decode. To address this, we propose PD-PRUNE, which identifies unpromising residuals during KV-cache encoding and defers continuation decisions to a short generated prefix. We introduce two unshared scores with ordered thresholds $(\tau_{\text{pre}}, \tau_{\text{dec}})$: the prefill score reads residual text and withholds repeated loops before the kernel and before transfer, while the decode score reads a probed prefix and extends a residual only when that prefix exceeds τ_{dec} . The two scores do not share weights; collapsing them onto a single threshold discards trajectories that remain solvable. Under disaggregation the connector inserts KV only for the admitted set. Experiments on Qwen3-8B show that PD-PRUNE reduces the prefill shared by three decode-time pruners while preserving any-finished accuracy, that a short probe restores full MATH-500 accuracy at lower end-to-end time, and that a three-turn Tree-of-Thoughts trace prefills 48 of 192 residuals.

1 Introduction

Agentic inference expands a committed prefix into multiple candidate residuals and subsequently retains a single trajectory (Yao et al., 2023a,b; Wang et al., 2023). This search-style generation has become standard for Tree-of-Thoughts, self-consistency, and tool-using loops. The resulting accuracy gains, however, come at a substantial serving cost: every residual incurs GPU prefill, KV transfer on a disaggregated connector, and decode until a pruning signal becomes available.

Why prune before decode? Early-stopping self-consistency (ESC), speculative rejection, and dynamic parallel tree search (DPTS) terminate unpromising paths once generated tokens can be scored (Li et al., 2024; Sun et al., 2024; Ding et al., 2025). As illustrated in Figure 1(a), this policy still encodes every residual, so three hosts share one prefill. Automatic prefix caching records a block only after it is full (Kwon et al., 2023; Zheng et al., 2024), and a disaggregated connector forwards every page the prefiler has computed (Zhong et al., 2024; Patel et al., 2024). Section 2 places each of these decisions on the fan-out. A single threshold on both prefill and decode can discard residuals that remain solvable once a prefix is observed. Admission therefore reads residual text, and continuation reads a generated prefix.

PD-PRUNE We introduce PD-PRUNE, which decouples the two decisions with ordered thresholds $(\tau_{\text{pre}}, \tau_{\text{dec}})$, $\tau_{\text{pre}} < \tau_{\text{dec}}$. The prefill score is computed from the residual and withholds a repeated loop before any kernel or KV transfer, so the disaggregated connector inserts only the admitted set. The decode score is computed from a generated prefix of length t_0 and extends a residual to the budget D only when that prefix exceeds τ_{dec} . The designated residual ρ_0 is always admitted, so pruning cannot erase the trajectory retained when no sibling survives. We show that the split is host-invariant in prefill work: any decode-time pruner that scores only generated tokens shares one W , equal to the prefill of the admitted set.

The inference pipeline comprises three stages—admission, prefill, and probe-and-resume—shown in Figure 2. PD-PRUNE leaves the host’s decode cut unchanged and introduces no additional model calls at test time.

Contributions. Our main contributions are:

- We identify a placement gap: search, decode-

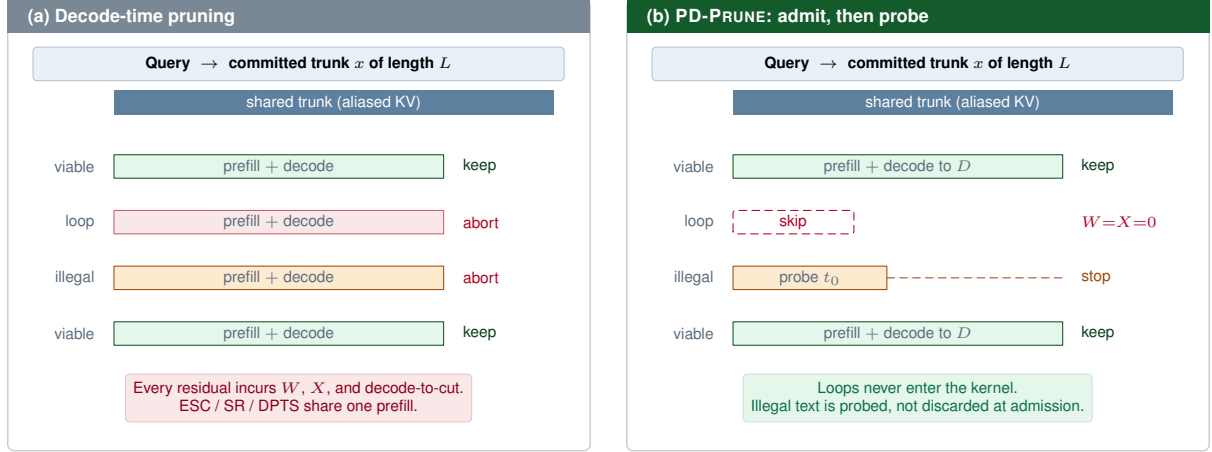


Figure 1: The necessity of disaggregated prefill and decode pruning. (a) Decode-time hosts score generated tokens, so every residual incurs prefill W , transfer X , and decode-to-cut before abort (Li et al., 2024; Sun et al., 2024; Ding et al., 2025). (b) PD-PRUNE withholds repeated loops before the kernel and continues an uncertain residual only after a short probe. Unpromising prefixes are terminated early, reducing compute without replacing the host.

time pruning, prefix caching, and prefill-decode disaggregation all decide after tokens or pages exist, so three hosts with different cuts share one prefill and one transfer.

- We propose PD-PRUNE, disaggregated prefill and decode pruning with unshared thresholds (τ_{pre} , τ_{dec}). The prefill score gates KV encoding and the connector; the decode score gates a short probe.
- We provide measurements showing that the prefill gate reduces shared prefill in front of ESC, speculative rejection, and DPTS while preserving any-finished accuracy, that a short probe restores full MATH-500 accuracy at lower end-to-end time, and that a three-turn Tree-of-Thoughts trace prefills 48 of 192 residuals.

2 Background

Agentic inference is a branching execution. A parent holds a committed trunk x of length L . The harness proposes k residuals ρ_i of length ℓ_i , and after the step a single spine $x \circ \rho_*$ remains. Tree-of-Thoughts makes that step explicit: a propose prompt writes candidate thoughts, and a value prompt then labels each thought sure, maybe, or impossible (Yao et al., 2023a). The value score is a function of a thought that has already been written. Chain-of-thought, self-consistency, and ReAct share the same order: expand, then score, then retain one trajectory (Wei et al., 2022; Wang et al., 2023; Yao et al., 2023b).

Where the prune score is read. Decode-time pruners move the cut earlier, and the score is still a function of generated tokens. Early-stopping self-consistency waits until a window of complete answers agrees (Li et al., 2024). On a window of length 5 that has not yet closed, every sample has been prefilled and decoded toward the budget. Speculative rejection spends a partial-reward horizon of 256 tokens and then drops the lower fraction of the batch (Sun et al., 2024). The prefill and the first 256 tokens are already resident when the quantile is applied. Dynamic parallel tree search generates a fixed mini-step on every active child and early-stops a child whose confidence is low (Ding et al., 2025). In our reproduction that mini-step is 100 tokens. An earlier cut inside this family shortens decode. It leaves the prefill unchanged, which is why ESC, speculative rejection, and DPTS share one prefill column when they run alone (Table 2).

Serving the same fan-out. Autoregressive serving splits that fan-out across two phases (Yu et al., 2022; Zhong et al., 2024; Patel et al., 2024). Prefill consumes the prompt in parallel and is compute-bound. Decode emits one token per step and is memory-bandwidth bound. PagedAttention stores KV in non-contiguous pages, and automatic prefix caching records a block only after the block is full (Kwon et al., 2023). RadixAttention indexes the same full prefix (Zheng et al., 2024). On the first fan-out those blocks are unpublished, so the engine materializes k copies of the trunk,

$$W_{\text{APC}} = kL + \sum_i \ell_i.$$

Method	Score input	First expand	Connector
ToT value	written thought	every residual	—
ESC, SR, DPTS	generated prefix	shared W	every child
APC, radix	full block	k copies of L	none
DistServe	none	k copies of L	(1)
PD-PRUNE	residual, then t_0	admitted set	miss of Keep

Table 1: Where one fan-out of k residuals pays. “Score input” is the earliest text the method can read. The PD-PRUNE row is the rule in Section 3: a repeated loop is withheld before the kernel, and the connector inserts only the admitted miss.

A later replay of the winner is a hit. Prefill-decode disaggregation isolates time-to-first-token from inter-token latency by placing the two phases on separate instances (Zhong et al., 2024; Patel et al., 2024). The connector then forwards every page the prefiller has computed. There is no admission on that path, so the transfer equals the first-expand prefill, including residuals the search will abort:

$$X_{PD} = kL + \sum_i \ell_i. \quad (1)$$

Table 1 records the stage at which each method can refuse a residual.

What the split has to separate. The same surface string is read twice. As a residual it is a candidate for KV encoding. As a generated prefix it is a candidate for continuation. A loop falls below both thresholds. Illegal math text can sit above the prefill threshold and below the decode threshold, so one shared cutoff either withholds a residual that can still solve or extends a prefix the decode rule is designed to stop. Retraining the model to emit fewer thoughts requires a heavier budget than a pair of string scores, and it leaves the connector unchanged. These two readings are why admission and continuation use ordered thresholds, and why the disaggregated insert follows the admitted set.

3 PD-PRUNE

3.1 Preliminary

Agentic prompting. Let $x = [x_t]_{t=1}^L$ denote a committed trunk of length L , and let $\rho_0, \dots, \rho_{k-1}$ denote k residuals proposed from x . Each candidate is the concatenation $z_i = x \circ \rho_i$. After the step, a single spine $x \circ \rho_*$ remains. Let $p_\theta(\cdot | z_i)$ be the next-token distribution of an LLM parameterized by θ . The model is expected to produce a correct answer from at least one viable residual; unpromising residuals should contribute neither prefill nor transfer.

State-of-the-art LLMs adopt the Transformer architecture (Vaswani et al., 2017). Let H_i denote the KV cache of z_i after prefill. For a continuation $y_i = [y_{i,t}]_{t=1}^D$ of budget D ,

$$p_\theta(y_{i,t} | z_i, y_{i,<t}) = p(y_{i,t} | H_i, y_{i,<t}, \theta). \quad (2)$$

H_i is reused during decode. On a copy-on-write tree, x is aliased in $O(1)$ and only ρ_i allocates new pages; live occupancy after abort is the spine.

Fan-out cost. We write W for the tokens actually executed by the prefill kernel, X for the KV transferred from a disaggregated prefill instance to a decode instance (Zhong et al., 2024; Patel et al., 2024), and decode-to-cut for the prefix generated before a prune score exists. On the systems in Figure 1(a), all three quantities are incurred before abort.

Decode-time pruning. ESC, speculative rejection, and DPTS assign scores to tokens that are already resident (Li et al., 2024; Sun et al., 2024; Ding et al., 2025). We write y_i for a continuation of z_i under a host cut: ESC decodes to D while an agreement window is open, speculative rejection scores at a partial-reward horizon, and DPTS generates a mini-step. A residual is wasted if H_i is computed and y_i is later aborted. Because y_i exists only after H_i is encoded, W is invariant to the host (Proposition 1).

3.2 Disaggregated Prefill and Decode Pruning

To reduce this waste, PD-PRUNE decouples admission from continuation. The inference process comprises three stages (Figure 2): scoring residual text and forming an admission mask (**Admit**), encoding KV only for the admitted set (**Prefill**), and generating a short prefix whose score determines whether the remaining budget is spent (**Probe and Resume**).

3.2.1 Stage I: Admission

Let $s^{\text{pre}} : \rho \mapsto [0, 1]$ be a prefill score on residual text. We score only ρ_i , not the trunk x , because unpromising content is proposed in the residual. Details of s^{pre} are deferred to Section 3.3.

The designated residual ρ_0 is retained whenever no sibling survives. Admission is therefore

$$\text{keep}_i = [i = 0 \vee s^{\text{pre}}(\rho_i) \geq \tau_{\text{pre}}]. \quad (3)$$

Only residuals with keep_i are queued for the kernel. W is the measured prefill of that set; a withheld residual contributes neither W nor X .

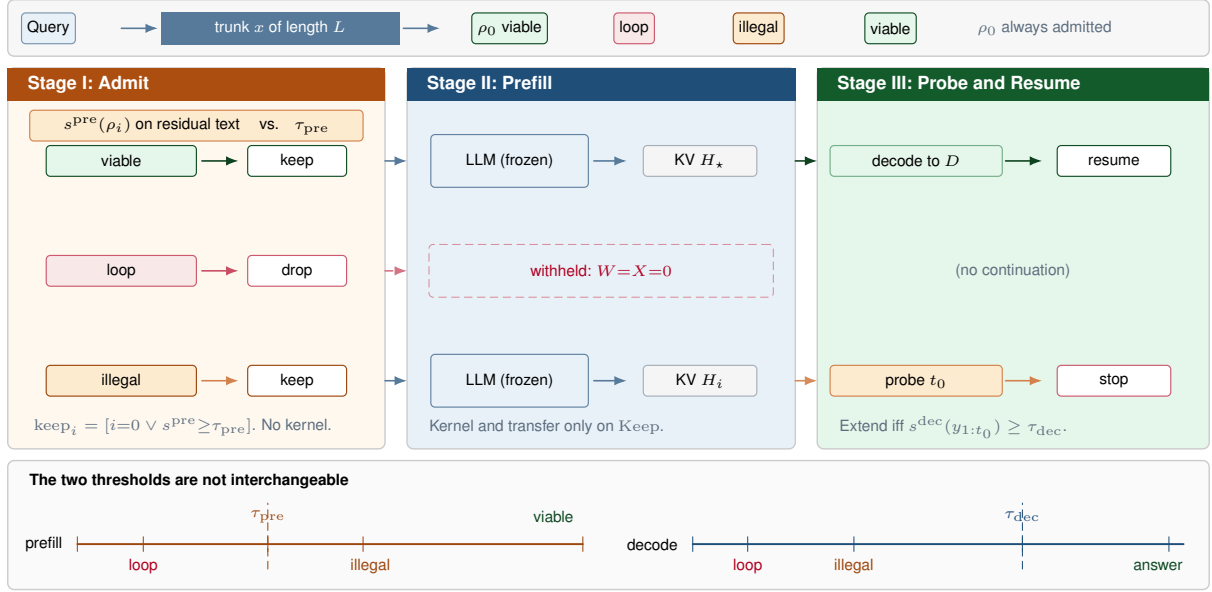


Figure 2: Inference pipeline of PD-PRUNE. The process comprises three stages: scoring residual text and forming an admission mask (**Admit**), encoding KV only for the admitted set (**Prefill**), and generating a short prefix whose score determines whether the remaining budget is spent (**Probe and Resume**). The two number lines show that illegal text lies above τ_{pre} on the residual and below τ_{dec} on a still-illegal prefix, so the thresholds are not interchangeable.

Applying τ_{dec} as a prefill threshold would discard illegal text that can still solve once a prefix is observed. The default policy therefore uses τ_{pre} for admission and reserves τ_{dec} for continuation. Raising the prefill threshold to τ_{dec} is a faster operating point that we measure, not the default.

3.2.2 Stage II: Prefill and Disaggregated Insert

Admitted residuals are encoded on the aliased trunk. The trunk is one shared prefix, so L is not replicated onto each residual, and prefix caching still applies inside the admitted set. The gate determines who is submitted. Baseline disaggregation forwards every computed page, which is the payload in (1). PD-PRUNE inserts KV only for $Keep = \{i : keep_i\}$:

$$X_i = W_i [i \in Keep]. \quad (4)$$

A withheld loop contributes neither prefill nor transfer. Admitted residuals that Stage III later stops have already incurred prefill; they do not incur the remainder of D , and pages that were never queued do not cross the connector. On the accounting model in Appendix H—ten sessions, $k=4$, one loop among four residuals—(4) reduces the connector from 11520 tokens to 8640.

Algorithm 1 Probe and resume. Loops skip prefill. ρ_0 decodes to D .

Require: residuals $\rho_{0:k-1}$, budget D , probe t_0 , thresholds (τ_{pre}, τ_{dec})

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1: for  $i \leftarrow 0$  to  $k-1$  do
2:   if  $i \neq 0$  and  $\rho_i$  is a repeated loop then
3:     skip  $\triangleright$  no kernel,  $X_i=0$ 
4:   else if  $i = 0$  or prior of  $\rho_i \geq \tau_{dec}$  then
5:     prefill  $\rho_i$ ; decode to  $D$ 
6:   else
7:     prefill  $\rho_i$ ; decode  $t_0$  tokens
8:     if  $s^{dec}(y_{i,1:t_0}) \geq \tau_{dec}$  then
9:       extend from  $t_0$  to  $D$ 
10:    end if
11:  end if
12: end for

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3.2.3 Stage III: Probe and Resume

Decode length is assigned after admission. A repeated loop is never probed. ρ_0 and any residual whose planner prior already exceeds τ_{dec} decode to D . Every other admitted residual generates t_0 tokens and is extended iff

$$extend_i = [s^{dec}(y_{i,1:t_0}) \geq \tau_{dec}]. \quad (5)$$

Algorithm 1 implements this rule. When a decode-time host is present (Section 3.4), the host supplies the later cut in place of t_0 .

3.3 Scoring Functions

Admission is guided by s^{pre} and continuation by s^{dec} . Both are deterministic functions of a string. They do not call the model and do not share weights.

The prefill scorer inspects ρ_i . A repeated span is a loop and falls below τ_{pre} . A residual matching an illegal-math pattern, such as a run of stars or an undefined token, lies strictly between τ_{pre} and τ_{dec} . Any other non-empty residual receives

$$s^{\text{pre}}(\rho_i) = 0.55 \min(1, 2\alpha_i) + 0.45 \min(1, 8\eta_i), \quad (6)$$

where α_i is the alphanumeric fraction and η_i is character-level uniqueness. Viable strategy text exceeds both thresholds.

The decode scorer inspects a generated prefix y_i . A repeated prefix stays below τ_{dec} . An answer mark (#### or \boxed) exceeds τ_{dec} . Illegal text on a prefix also stays below τ_{dec} , which is why it cannot share τ_{pre} . Otherwise

$$s^{\text{dec}}(y_i) = 0.50 \min(1, 2\alpha) + 0.30 \min(1, 8\eta) + 0.20 \min(1, 20\delta + 15\omega), \quad (7)$$

with δ the digit fraction and ω the fraction of arithmetic symbols. A static planner prior used by the host-side ranker is a third mix (0.35 alphanumeric, 0.65 uniqueness) and is never substituted for s^{pre} .

The operating point is the ordered pair $(\tau_{\text{pre}}, \tau_{\text{dec}})$ with $\tau_{\text{pre}} < \tau_{\text{dec}}$ (Equation 8). On the prefill axis the loop lies below τ_{pre} and illegal text above it, so admission drops loops and keeps illegal text. On the decode axis illegal text lies below τ_{dec} and an answer mark above it, so a probe stops on illegal text and resumes once an answer mark appears. Section 5 measures the effect of using a single threshold for both decisions.

Score gap. The same illegal span can be interpreted as a residual to encode or as a prefix to stop, depending on whether the score is read before or after generation. We refer to this discrepancy as the *score gap*. Figure 2 plots the four characteristic points on two number lines. Applying τ_{dec} as a prefill threshold drops illegal residuals that can still solve once a prefix is visible; applying τ_{pre} as a decode threshold would extend an illegal prefix. Motivated by the gap, continuation uses only a prefix of length t_0 . On MATH-500, $t_0=128$ is the setting that returns $k=16$ any-finished accuracy to the full fan-out.

Proposition 1 (Host-invariant prefill). *Let $\text{Keep} = \{i : \text{keep}_i\}$ be the admitted set of (3). For any decode-time host that scores only generated tokens, the prefill work W equals the measured prefill of Keep and does not depend on the host.*

The proof is in Appendix C. Proposition 1 explains the identical prefill column in Figure 1(a) and Table 2.

3.4 Compatibility with Decode-Time Hosts

ESC, speculative rejection, and DPTS remain the decode policy. PD-PRUNE changes only the set of residuals that reach them.

- ESC decodes an admitted residual to D while its agreement window is open. On the evaluation mix the window stays open, so an admitted residual runs to $D=512$.
- Speculative rejection scores at horizon $\tau=256$ and drops the lower fraction $\alpha=0.5$ of the admitted set.
- DPTS generates a mini-step of 100 tokens and early-stops a child whose confidence is below τ_{dec} .

The pair $(\tau_{\text{pre}}, \tau_{\text{dec}})$ is not retuned per host. When the prefill threshold is raised to τ_{dec} , the admitted set is the viable half, the host-side drop count falls to zero, and the three hosts converge to the same decode length. PD-PRUNE is complementary to these hosts: it does not modify their scoring rules. A batch-limited rule that ignores the score and retains a single residual is a different policy; Section 5 shows that it discards correct trajectories.

4 Related Work

Path pruning during search. Chain-of-thought prompting showed the value of intermediate steps (Wei et al., 2022). Tree-of-Thoughts, self-consistency, and ReAct expand several reasoning paths and then retain one trajectory (Yao et al., 2023a; Wang et al., 2023; Yao et al., 2023b). A later line prunes those paths once they have been written. EquivPruner trains a lightweight detector of mathematical statement equivalence and keeps one representative from each equivalent set, cutting token use during search (Liu et al., 2026). STOP reads the KV cache of a generated prefix through a learned token and resumes only the top-ranked paths, which is a launch-check-resume schedule

over prefixes that already exist (Bi et al., 2026). Both methods shorten exploration after the candidate text is resident. An earlier threshold inside that family reduces decode and leaves the prefill of every sibling in place. PD-PRUNE reads the residual before those tokens exist, inserts KV only for the admitted set, and leaves the host cut in place for residuals that remain. EquivPruner and STOP can still rank that admitted set: they decide which surviving residual to continue, which is the role of s^{dec} and of the host.

Decode-time early stopping. ESC closes the sample once a window of answers agrees (Li et al., 2024); speculative rejection spends a partial-reward horizon and then drops the lower quantile (Sun et al., 2024); DPTS grows a tree in mini-steps and stops a child whose confidence is low (Ding et al., 2025). Section 2 records the text each score reads. The decode score in PD-PRUNE is a probe of a sibling residual, whereas speculative decoding checks a draft continuation of the same sequence (Leviathan et al., 2023).

LLM serving. Orca batches at iteration granularity (Yu et al., 2022). PagedAttention and automatic prefix caching address KV as blocks and reuse a block after its hash is stored (Kwon et al., 2023); SGLang’s radix cache implements the same contract for program-structured calls (Zheng et al., 2024). DistServe and Splitwise place prefill and decode on separate instances and ship KV across the cut (Zhong et al., 2024; Patel et al., 2024). These designs remove recomputation of a stored prefix and isolate time-to-first-token from inter-token latency. The first fan-out of a new trunk has nothing stored, and the connector forwards every page the prefiller computed, including residuals the harness will abort. Training-time methods that modify the model, and test-time workflows that add extra model calls per response, require a larger budget than a pair of string scores. PD-PRUNE is the admission step in front of that kernel and that transfer. It is complementary to the existing serving stack and uses the page allocator those systems already provide.

5 Experiments

In this section we present experiments designed to validate the efficacy of PD-PRUNE.

5.1 Experimental Setup

We evaluate on Qwen3-8B at temperature 0 with prefix caching and tensor parallelism 1, on four A100 40GB GPUs. By default we use the ordered pair in Equation 8 and probe length $t_0=128$. We evaluate GSM8K under three decode-time hosts and MATH-500 under the probe rule. We further report peak KV, quality, and offered load on a serving stack whose live occupancy is the committed spine, using six model families. Host parameters and metric definitions are given in Appendix A. PD-PRUNE is complementary to the hosts: it does not modify their scoring rules or introduce additional test-time model calls.

GSM8K hosts. Sixteen GSM8K problems, each a fan-out of $k \in \{4, 8, 16\}$ residuals with decode budget $D=512$. Each fan-out is a controlled mix: half the residuals are viable strategy instructions, and half alternate a repeated loop with illegal text. ρ_0 is always a viable strategy. *Any-finished* counts a problem if any admitted residual produces the reference number. *Winner* counts a problem if the residual selected as the winner does. Both are reported over 16. ESC, speculative rejection, and DPTS are the decode hosts. The host alone submits every residual; PD-PRUNE applies τ_{pre} first and leaves the host cut in place.

MATH-500 probe. The first 32 MATH-500 problems at level 4 or higher, $k \in \{4, 8, 16\}$, $D=1024$. Half the residuals are contest strategies and half are loops or illegal text. The decode threshold is τ_{dec} . Full fan-out decodes every residual to D . Discard drops every residual below that threshold before prefill. Probe admits those residuals for t_0 tokens and extends only when the generated prefix exceeds τ_{dec} .

5.2 Results

Table 2 and Figure 3 summarize the main findings. PD-PRUNE reduces the prefill shared by three decode-time hosts while preserving any-finished accuracy.

Specifically, with the prefill gate disabled, the three hosts share one prefill of 5008 tokens. Enabling τ_{pre} withholds loops, retains illegal text, and reduces that prefill to 3848 on every host. ESC exhibits the largest time reduction because its baseline decodes loops to D . Speculative rejection and DPTS improve less, because they already terminate the unpromising half at their own cut. The gate re-

$k=16$	Prefill	Host	$+\tau_{\text{pre}}$	Any
ESC	5008 $\rightarrow$ 3848	17.05 s	14.13 s	16/16
SR	5008 $\rightarrow$ 3848	13.14 s	12.47 s	16/16
DPTS	5008 $\rightarrow$ 3848	11.86 s	11.49 s	16/16

Table 2: Prefill pruning at τ_{pre} in front of three hosts. Any-finished stays 16/16. The prefill column is identical across hosts because the gate, not the host cut, changes W , as predicted by Proposition 1.

moves loop prefill, which none of those cuts avoids. This is a limitation of pruning only after tokens exist: the host can stop an unpromising residual, but it cannot recover W .

Figure 3(a) raises the prefill threshold from τ_{pre} to τ_{dec} , which also withholds illegal text. Any-finished holds at $k=16$. End-to-end time converges across the three hosts, and the host-side cut is never invoked: admission has already removed every residual that cut would have rejected.

Table 3 isolates the two stages. Decode pruning alone never changes prefill, and the time ranking is the ranking of the cut. Prefill pruning alone, with no host cut, is already faster than ESC at τ_{pre} and faster than all three hosts when the prefill threshold is τ_{dec} . Stacking a host on τ_{pre} still stops some admitted residuals. Stacking a host on τ_{dec} does not, because the admitted set is the viable half. Any-finished holds throughout. Winner accuracy is not identical to any-finished: withholding illegal residuals can change which residual the host ranks first. The gate preserves a correct residual in the admitted set. Appendix E reports $k=4$ and $k=8$, where the higher prefill threshold does reduce any-finished accuracy.

The two scores are ordered for a reason (Section 3.3). Illegal text lies above τ_{pre} on the residual and below τ_{dec} on a prefix that is still illegal. Using τ_{dec} as the prefill threshold therefore withholds residuals that (3) would admit. On MATH-500 that choice is the discard row of Table 4.

Table 4 and Figure 3(b) compare the three prefill policies on MATH-500. Discarding every residual below τ_{dec} is the fastest curve and the least accurate. The probe spends t_0 tokens on uncertain residuals, then extends a residual only when its prefix exceeds τ_{dec} . At $k=16$ any-finished returns to the full fan-out. At $k=4$ and $k=8$ it does not. Recovery is a wide-fan-out result. Appendix F sweeps $t_0 \in \{64, 128\}$, loop-skipping, and a decode-only short budget. Among the probe lengths we compare, the $k=16$ match to full accuracy is specific to $t_0=128$;

$t_0=64$ reaches 18/32.

5.3 Disaggregated Search Traces

The host experiments score one fan-out. A search trace is a sequence of fan-outs on a committed spine. We run three-turn Tree-of-Thoughts on Qwen3-4B and Qwen3-8B at $\text{tp}=2$, $n=16$, $k=4$, temperature 0, one problem per generate. Turns 1 and 2 decode a 32-token step on the children admission kept. The designated residual becomes the parent of the next turn. Turn 3 answers on that spine, with $D=512$ on GSM8K and $D=256$ on Game24. Across 16 problems the trace offers 192 residuals. PD-PRUNE queues the designated residual each turn and withholds the other three before the kernel, so 48 residuals are prefilled.

	Qwen3-4B		Qwen3-8B	
	APC	PD-PRUNE	APC	PD-PRUNE
GSM8K e2e (s)	1.66	1.63	2.48	1.91
GSM8K acc.	15/16	15/16	16/16	15/16
Game24 e2e (s)	1.61	1.13	2.15	1.52
Game24 acc.	6/16	9/16	9/16	10/16
GSM8K expand (ms)	429	55	597	58
Game24 expand (ms)	435	53	598	56

Table 5: Three-turn Tree-of-Thoughts, $n=16$, $k=4$, $\text{tp}=2$. End-to-end time is seconds per problem. Expand is the prefill of the k residuals across three turns, in milliseconds per problem. PD-PRUNE prefills 48 of 192 residuals. Expand is 7.8–10.7 $\times$ below automatic prefix caching.

Table 5 separates that expand from the winner decode. On Qwen3-8B, GSM8K end-to-end time falls from 2.48 s to 1.91 s and Game24 from 2.15 s to 1.52 s. GSM8K accuracy is unchanged on Qwen3-4B and one problem lower on Qwen3-8B. Game24 accuracy rises on both models. Peak KV on the live spine is 275 tokens against 318 for prefix caching on GSM8K, and 216 against 265 on Game24, because abort leaves the committed path. Table 6 is a different decision inside one uncommitted fan-out: retaining a single slot there drops siblings that still solve the problem. The insert itself is (4). On the loop-only accounting model, withholding one residual per session reduces the connector from 11520 tokens to 8640 (Appendix H).

The prefill gate does not lengthen the committed spine. Appendix K measures that spine against recomputation and automatic prefix caching. Peak KV stays below the cache on six model families, because live occupancy is the spine. On Qwen3-8B the pruned system keeps GSM8K quality with that cache and lowers end-to-end time. Under a 1 s P99

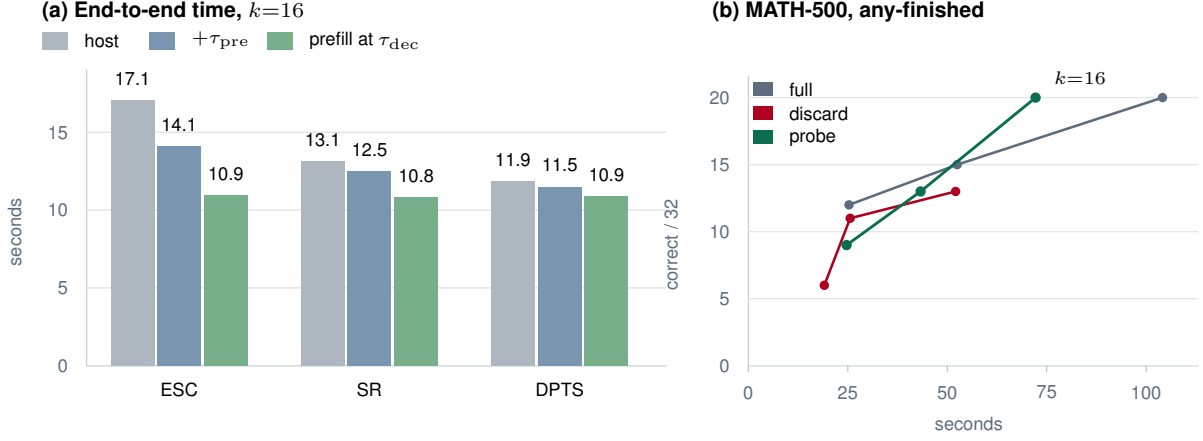


Figure 3: Qwen3-8B. (a) GSM8K controlled mix, $k=16$, $n=16$, $D=512$. Each group is one host. Gray denotes the host alone, blue adds τ_{pre} , and green applies τ_{dec} at prefill. Any-finished accuracy is unchanged in every setting. At the higher prefill threshold the three hosts coincide. (b) MATH-500, $n=32$, $D=1024$. Each polyline connects $k=4$, $k=8$, and $k=16$. Discarding residuals below τ_{dec} before prefill is fast and less accurate. The probe under (τ_{pre}, τ_{dec}) meets the full fan-out at $k=16$.

Setting	Admit	Cut later	Prefill	Decode	Time	Any	Winner
<i>Decode pruning only.</i> No τ_{pre} . Every residual is prefilled.							
ESC, budget $D=512$	16	0	5008	131072	17.05 s	16/16	11/16
SR, horizon 256	16	8	5008	98304	13.14 s	16/16	12/16
DPTS, mini-step 100	16	8	5008	78336	11.86 s	16/16	12/16
<i>Prefill pruning only.</i> Decode runs each admitted residual to $D=512$.							
τ_{pre}	12	0	3848	98304	14.13 s	16/16	11/16
τ_{dec} as prefill	8	0	2768	65536	10.92 s	16/16	10/16
<i>Both.</i> The same prefill threshold, then the host cut on what remains.							
SR + τ_{pre}	12	4	3848	81920	12.47 s	16/16	10/16
DPTS + τ_{pre}	12	4	3848	71936	11.49 s	16/16	12/16
SR + τ_{dec}	8	0	2768	65536	10.80 s	16/16	10/16
DPTS + τ_{dec}	8	0	2768	65536	10.90 s	16/16	10/16

Table 3: Ablation at $k=16$ on the GSM8K controlled mix. “Cut later” is the number of admitted residuals the host stops at its own horizon. Any-finished is unchanged. Winner accuracy moves by at most two problems. When the prefill threshold is τ_{dec} , the host cut is never invoked and the decode columns coincide.

	$k=4$	$k=8$	$k=16$
Full	12/32, 25.3 s	15/32, 52.5 s	20/32, 104.1 s
Discard	6/32, 19.1 s	11/32, 25.6 s	13/32, 52.1 s
Probe	9/32, 24.7 s	13/32, 43.3 s	20/32, 72.2 s

Table 4: MATH-500, Qwen3-8B, $n=32$, $D=1024$. Each cell is any-finished accuracy and end-to-end time. At $k=16$ the probe matches the full fan-out and is 31% faster. At $k=4$ and $k=8$ it does not fully recover accuracy.

$k=16$, threshold τ_{dec}	Admit	Prefill	ESC	SR
Keep one, ignore score	1	414	11/16	9/16
Keep all above τ_{dec}	8	2768	16/16	16/16

Table 6: Winner-only admission on the GSM8K mix. A single reserved slot reduces any-finished accuracy on both hosts. Scoring, and retaining the viable half, restores 16/16.

Table 6 holds the batch width fixed and compares two admission rules at τ_{dec} . Ignoring the score and retaining one residual reduces any-finished accuracy on both hosts. Retaining every residual at or above τ_{dec} restores it. The savings in Table 2 come from withholding loops, and at τ_{dec} from withholding illegal text as well. They do not come from collapsing the fan-out onto its winner.

time-to-first-token line, the shorter spine admits a higher offered load.

6 Discussion

We presented a lightweight approach to efficient agentic inference. By identifying unpromising residuals during KV-cache encoding, and by resuming an uncertain residual only when a short prefix exceeds a second threshold, PD-PRUNE ensures that prefill and transfer are spent on viable strategies rather than on text that search will abort. Experiments show a host-invariant reduction in shared prefill, lower end-to-end time, and preserved large-fan-out any-finished accuracy on GSM8K and MATH-500. On a three-turn Tree-of-Thoughts trace the same gate prefills 48 of 192 residuals and shortens expand by $7.8\text{--}10.7\times$ relative to prefix caching. Collapsing either decision onto the other threshold, or onto a single reserved slot inside an uncommitted fan-out, is what loses problems. The split is complementary to decode-time hosts, prefix caches, and prefill-decode disaggregation: the connector inserts only the admitted miss.

7 Limitations

The scores are surface tests: repetition, an illegal-math pattern, an answer mark, and character statistics. They are not a learned reward model, and a residual that is wrong without being repetitive passes the prefill gate. The evaluation mix is constructed so that half the fan-out is viable strategy text and half is a loop or illegal string. It is a controlled test of the two thresholds, not a trace of a production agent. Any-finished accuracy can stay flat while winner accuracy moves, and at $k=4$ even τ_{pre} drops one GSM8K problem. The MATH-500 probe recovers full any-finished accuracy at $k=16$ and does not do so at $k=4$ or $k=8$. The pruning study uses one model and temperature 0, so the tables are single-run measurements. The connector numbers in Appendix H are an accounting model, not a measured transfer time. The three-turn trace admits the designated residual after the spine is committed, so it does not replace a learned value model, and Qwen3-8B GSM8K accuracy moves by one problem. We do not explore training-based alternatives such as finetuning a pruner, since we target the setting without a heavy compute budget.

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A Additional Details

Metrics. We evaluate GSM8K with two accuracy numbers and two cost numbers. Any-finished $\uparrow$ is the fraction of problems for which any admitted residual produces the reference number. Winner $\uparrow$ is the fraction for which the residual selected as the winner does. Prefill $\downarrow$ is the measured GPU prefill token total. Time $\downarrow$ is end-to-end wall-clock seconds on four A100 40GB GPUs. For MATH-500 we report any-finished and time on the first 32 problems at level 4 or higher. Peak KV is the maximum live bytes of key-value cache; offered load is the concurrent request rate under a 1 s P99 time-to-first-token line.

Hosts. ESC uses an agreement window of length 5. On the controlled mix the window does not fill, so an admitted residual runs to $D=512$. Speculative rejection uses horizon 256 and dropped fraction 0.5, applied to the admitted set. DPTS uses a mini-step of 100 tokens. The memory-limit early trigger from speculative rejection does not activate at $k \leq 16$ on this hardware.

Admission rules. The default rule keeps every residual at or above τ_{pre} , including index 0. Winner-only keeps a single reserved slot and ignores the score. Both are measured at a fixed batch width in Table 6. PD-PRUNE is implemented on top of the host-alone baseline. It is complementary to the hosts, since it does not modify their scoring rules.

Controlled mix. For each problem, half of the residuals are viable strategy instructions and half alternate a repeated loop with illegal text. ρ_0 is always a viable strategy. The mix is a controlled test of the two thresholds, not a production trace. Worked strings are in Appendix J.

B Score Functions

The runs in this paper use one ordered pair,

$$(\tau_{\text{pre}}, \tau_{\text{dec}}) = (0.15, 0.45). \quad (8)$$

A loop scores 0.02 on the residual and 0.04 on a prefix. Illegal text scores 0.22 on the residual and 0.16 on a prefix. An answer mark scores 0.92. Thus τ_{pre} sits between a loop and illegal text, and τ_{dec} sits between an illegal prefix and an answer mark. Raising the prefill threshold to τ_{dec} is the faster operating point reported in the tables.

Both scorers are deterministic functions of a string. They do not call the model. Table 7 evaluates the residual kinds in the controlled mix.

Text	Reason	Prefill	Decode
Repeated <code>*loop* span</code>	loop	0.02	0.04
undefined nan ...	illegal	0.22	0.16
Viable strategy instruction	ok	≈ 1.00	—
Prefix containing <code>####</code> or <code>\boxed</code>	answer	—	0.92
Empty string	empty	0.00	0.00

Table 7: Scores on the residual kinds in the evaluation mix. A dash means that scorer is not applied to that string. The decode column is the score of a generated prefix of that kind, not a second pass over the residual.

Prefill. After the loop and illegal tests, (6) mixes an alphanumeric fraction with character uniqueness. The coefficients 0.55 and 0.45 are fixed. The measurements score the residual text.

Decode. The answer-mark test runs before the illegal test, so a prefix that both looks noisy and contains an answer mark is an answer (0.92). The numeric mix in (7) adds digit and operator mass so that a healthy math prefix can continue before an answer mark appears. The planner prior that decides who is probed is

$$0.35 \min(1, 2\alpha) + 0.65 \min(1, 8\eta)$$

on the residual, with loop 0.04 and illegal 0.16. It ranks residuals for the host. It is not substituted for s^{pre} .

Early prefix inside prefill. A separate test, not used by the tables in Section 5, inspects the first fraction $\alpha=0.20$ of a residual. It triggers when that slice scores below τ_{pre} for a reason other than ordinary text. The reported measurements use the full-residual loop test rather than this partial abort.

C Proof of Proposition 1

Proposition (Host-invariant prefill). *Let $\text{Keep} = \{i : \text{keep}_i\}$ be the admitted set of (3). For any decode-time host that scores only generated tokens, the prefill work W equals the measured prefill of Keep and does not depend on the host.*

Proof. A decode-time host is a map from generated prefixes to a stop/continue decision (Li et al., 2024; Sun et al., 2024; Ding et al., 2025). By (2), a prefix $y_{i,t}$ exists only after the KV cache H_i of z_i has been encoded. Therefore the host cannot remove z_i from the prefill batch: its cut acts on decode length, not on whether ρ_i is queued. Equation (3) is computed from residual text before any kernel, so Keep is independent of the host. W is defined as the tokens the kernel runs on that set. Hence W is a function of Keep alone. The three hosts in Table 2 therefore share one prefill column at each threshold, and differ only in decode-to-cut. $\square$

The proposition does not claim that end-to-end time is host-invariant. Time still ranks by the cut: ESC to D , speculative rejection to horizon 256, DPTS to a mini-step of 100. Prefill pruning changes the time ranking only by changing Keep , which is the empirical content of Table 3.

D Token Accounting

The decode columns of Table 3 are exact products of the admission counts, the cut, $D=512$, and $n=16$ problems. Write a for the number of admitted residuals per problem and c for how many of those the host stops at horizon h , with the rest running to D . Decode tokens are $n(c \cdot h + (a - c) \cdot D)$.

- ESC, no prefill gate: $a=16, c=0, h$ unused. $16 \cdot 16 \cdot 512 = 131072$.
- Speculative rejection: $a=16, c=8, h=256$. $16 \cdot (8 \cdot 256 + 8 \cdot 512) = 98304$.
- DPTS: $a=16, c=8, h=100$. $16 \cdot (8 \cdot 100 + 8 \cdot 512) = 78336$.
- Prefill at τ_{pre} only: $a=12, c=0$. $16 \cdot 12 \cdot 512 = 98304$.
- Prefill at τ_{dec} only: $a=8, c=0$. $16 \cdot 8 \cdot 512 = 65536$.
- SR + τ_{pre} : $a=12, c=4, h=256$. $16 \cdot (4 \cdot 256 + 8 \cdot 512) = 81920$.

- DPTS + τ_{pre} : $a=12, c=4, h=100$. $16 \cdot (4 \cdot 100 + 8 \cdot 512) = 71936$.
- Either host + τ_{dec} : $a=8, c=0$. 65536.

The mix explains the admission counts. For $k=16$, eight residuals are viable and eight are not. That half alternates loop and illegal text, four of each. τ_{pre} withholds the four loops and retains the four illegal residuals: $8 + 4 = 12$. Raising the prefill threshold to τ_{dec} withholds both kinds: 8. ρ_0 is among the viable eight, so the forced keep does not add a further slot. At $k=8$ the same halving admits 6 at τ_{pre} and 4 at τ_{dec} . At $k=4$ it admits 3 and 2.

Prefill tokens are measured, not assigned by this product, because prefix caching makes the incremental prefill depend on the token ids of each residual. The measured totals at $k=16$ are 5008, 3848, and 2768 for admission 16, 12, and 8. The same totals appear for every host at a given τ_{pre} , which is the empirical content of Figure 1(b) and of Proposition 1.

E Fan-out Width

ESC	Base	+ τ_{pre}	Δ	Any	Winner	Prefill
$k=4$	8.65 s	8.22 s	-5%	16→15	9→9	1320→1018
$k=8$	10.84 s	10.19 s	-6%	16→16	11→9	2624→2020
$k=16$	17.05 s	14.13 s	-17%	16→16	11→11	5008→3848

Table 8: ESC across fan-out width at the default prefill threshold. The relative saving grows with k because more loop residuals are withheld. At $k=4$ any-finished drops by one problem. At $k=8$ any-finished holds and the winner moves by two.

	Prefill	Admit	ESC	Any
$k=16$ base	5008	16	17.05 s	16/16
$k=16, \tau_{\text{pre}}$	3848	12	14.13 s	16/16
$k=16, \tau_{\text{dec}}$ as prefill	2768	8	10.92 s	16/16
$k=8, \tau_{\text{dec}}$ as prefill	1448	4	—	14/16
$k=4, \tau_{\text{dec}}$ as prefill	732	2	—	14/16

Table 9: Why the default prefill threshold is τ_{pre} . At $k=16$, raising it to τ_{dec} is faster and any-finished still holds. At $k=8$ and $k=4$ the same threshold admits only the viable half, which is four and two residuals, and any-finished falls to 14/16. Speculative rejection at $k=16$ with that higher threshold is 10.80 s and DPTS is 10.90 s, both 16/16.

The $k=4$ row of Table 8 is the reason Section 7 does not describe τ_{pre} as accuracy-neutral at every width. One problem that the full ESC fan-out

solved is absent once the single loop residual at $k=4$ is withheld. We retain τ_{pre} because raising it to τ_{dec} , which looks better on the $k=16$ groups of Figure 3(a), fails the same test two widths down (Table 9).

F Probe Sweep

Two comparisons in Table 10 fix the design of Algorithm 1. Dropping loops and then decoding everyone else to D is the prefill gate with no probe. At $k=16$ it is slightly faster than the probe (71.4 s versus 72.2 s) and four problems less accurate. The accuracy belongs to the residuals that fail the prefill-side viability test and still solve after a prefix is visible. A short decode that never consults s^{dec} reaches 17/32: the extension test, not the mere existence of a short budget, is what returns the last problems. Including loops in the probe matches 20/32 at 74.4 s. We still skip loops before the kernel. Their prefill score is below τ_{pre} , the probe does not buy accuracy at $k=16$, and it spends the extra 2.2 s.

At $k=16$ and $t_0=128$, every non-loop uncertain residual is probed and then extended, and decode tokens fall from 524288 to 393216. The $k=8$ probe at $t_0=64$ reaches 16/32, one above the full fan-out’s 15/32. That is within the one-problem noise we decline to interpret; the claim we use in the body is the $k=16$ tie with the full fan-out at lower time.

G Host Cuts

The three hosts are unchanged except for the set (3) hands them.

ESC. The published stopping rule waits until a window of agreeing samples is full. The window length used here is 5. The controlled fan-out does not fill that window: some residuals are loops or illegal text, and the viable strategies do not agree early. Every admitted residual is therefore decoded to D . This is why ESC’s baseline is the most expensive row in Table 3, and why withholding loops before prefill reduces ESC more than it reduces DPTS.

Speculative rejection. The partial-reward horizon is $\tau=256$ tokens and the dropped fraction is $\alpha=0.5$, applied to the admitted set rather than to the original k . With 12 admitted residuals the host stops 4; with 8, all eight are above the mass the rule would reject, and the later-cut column is 0. The

memory-limit early trigger described in that paper does not activate at $k \leq 16$ on this hardware, so we do not take credit for it.

DPTS. Each admitted child generates a mini-step of 100 tokens. A child whose confidence is below τ_{dec} stops; the others continue to D . The mini-step is the earliest of the three cuts, which is why DPTS is the fastest host when the prefill gate is off (11.86 s) and why adding the gate moves it by only 0.37 s at τ_{pre} .

Order inside one problem. The prefill gate runs before any decode kernel for that fan-out. The host then runs on the children that were admitted. Withheld residuals are absent from the decode batch. The gates are ordered by cost: a text test, then prefill, then a short probe or the host horizon, then the rest of D . Only admitted KV is eligible to cross to the decode instance.

H Connector Accounting

The seconds in Section 5 are measured GPU time. The calculation below is a separate accounting model of the transfer term X .

Consider 10 sessions, $k=4$, trunk $L=256$, residual length $\ell=32$, with one of the four residuals a loop. A disaggregated prefill that has no admission computes every residual and ships every page:

$$W = X = 10 \cdot 4 \cdot (256 + 32) = 11520$$

tokens. Charging $12 \mu\text{s}$ per prefill token and $2 \mu\text{s}$ per transferred token, plus 0.8 ms to abort a withheld child, gives a fan-out of 162.1 ms of which 23.0 ms is transfer. Refusing the loop before insert leaves three residuals, each still carrying its trunk under this model:

$$W = X = 10 \cdot 3 \cdot 288 = 8640.$$

Transfer falls to 17.3 ms and the modeled fan-out to 121.8 ms. The 2880 tokens removed are one residual per session, trunk included, because this model replicates the trunk on the connector. A page cache that hits the trunk would shrink both the baseline and the pruned column by the same shared prefix; the residual term would remain. The model shows that the prefill gate and the connector payload move together once admission precedes the transfer. The measurements already include prefix caching, so the prefill column in Section 5 is incremental, not this 11520-token construction.

Setting	$k=4$	$k=8$	$k=16$
Full fan-out	12/32, 25.3 s	15/32, 52.5 s	20/32, 104.1 s
Discard every residual below τ_{dec} before prefill	6/32, 19.1 s	11/32, 25.6 s	13/32, 52.1 s
Drop loops only, decode the rest to D	11/32, 23.0 s	17/32, 41.5 s	16/32, 71.4 s
Short-decode the low scores, no extension test	10/32, 23.1 s	15/32, 42.3 s	17/32, 74.0 s
Probe $t_0=64$, extend iff prefix $\geq \tau_{\text{dec}}$	8/32, 24.0 s	16/32, 42.0 s	18/32, 72.5 s
Probe $t_0=128$, extend iff prefix $\geq \tau_{\text{dec}}$	9/32, 24.7 s	13/32, 43.3 s	20/32, 72.2 s
Probe loops as well ($t_0=128$)	10/32, 24.9 s	15/32, 43.3 s	20/32, 74.4 s

Table 10: MATH-500 ablation. Each cell is any-finished accuracy and wall-clock time. Dropping loops only is fast and, at $k=16$, loses four problems relative to the full fan-out (16/32 versus 20/32). Probing the non-loop residuals for 128 tokens and then applying τ_{dec} is the only row that matches 20/32. Probing the loops too matches accuracy and spends 2.2 s more. At $k=8$, several rows land within two problems of the full fan-out; we do not treat a one-problem difference on $n=32$ as a ranking.

I Winner-Only Admission

Table 6 uses batch size 8 on the $k=16$ GSM8K mix at threshold τ_{dec} . The score-blind rule fills one slot. Measured prefill is 414 tokens and any-finished accuracy is 11/16 under ESC and 9/16 under speculative rejection. The score rule admits the eight viable residuals (2768 prefill tokens) and both hosts return to 16/16. The gap is the viable strategies other than ρ_0 . Several problems are solved by a non-winner viable residual and by none of the single reserved slots the blind rule happened to retain. Retaining the argmax of the prefill score would be a third policy; we do not claim the prefill score ranks answer correctness, only that thresholding it is a different decision from retaining one residual.

J Worked Residuals

The viable half cycles through six instructions:

1. Translate the story into equations, then solve for the missing value.
2. Work backwards from the asked quantity to the given numbers.
3. Name each intermediate quantity in order and add them last.
4. Build a small table of given facts, then apply one operation at a time.
5. Estimate the magnitude first, then do exact arithmetic to confirm.
6. Convert all units, cancel common factors, then compute.

None of them matches the loop or illegal patterns, and (6) scores them at 1.0. The other half is the alternating pair

- a repeated `*loop*` span, scored 0.02 because a span of length 8 to 40 repeats at least three extra times;
- the string “undefined nan junk residual that cannot be a proof”, scored 0.22 by the illegal-math pattern.

A generated prefix is scored only after it exists. The illegal decode score and the answer score in Figure 2 are prefix scores. They are not a second labeling of the residual.

K Memory and Offered Load

The prefill gate does not lengthen the committed spine. Figures 4, 5, and 6 measure that spine against recomputation and automatic prefix caching on the same serving stack. Live occupancy is $b(L + \ell_*)$ after abort, so peak KV stays below automatic prefix caching.

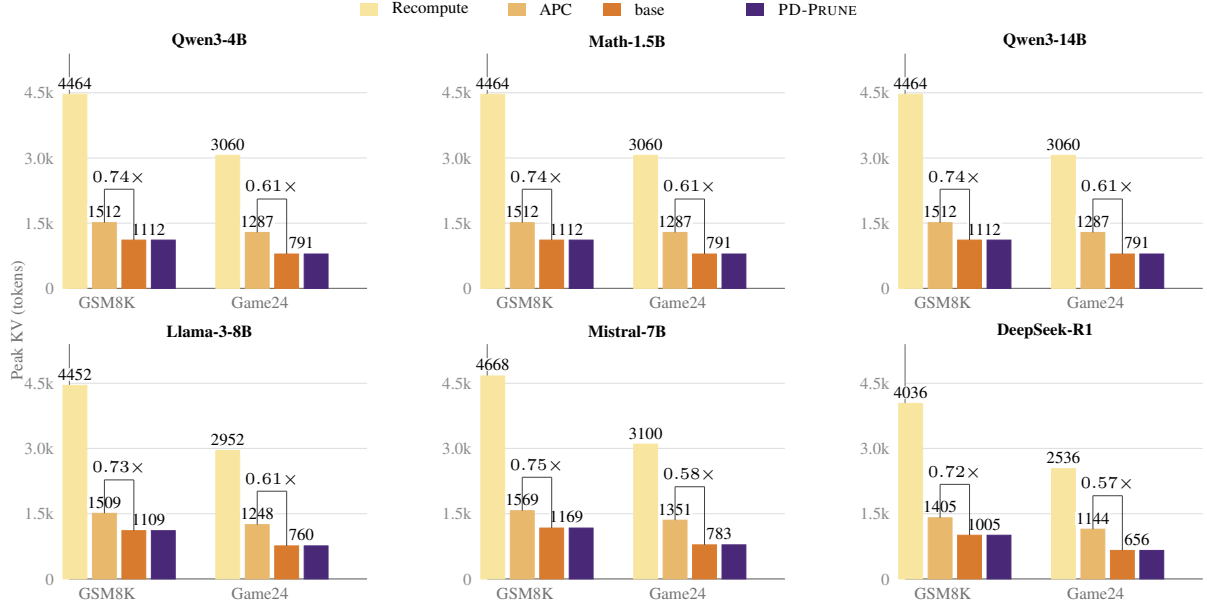


Figure 4: Peak KV on six models ($n=40$, $tp=2$). Each panel is one family; groups are GSM8K and Game24. Bars are recompute, APC, the unpruned spine, and PD-PRUNE. PD-PRUNE shares the spine with the unpruned runtime, so the two right-hand bars match. The annotated ratio is spine over APC.

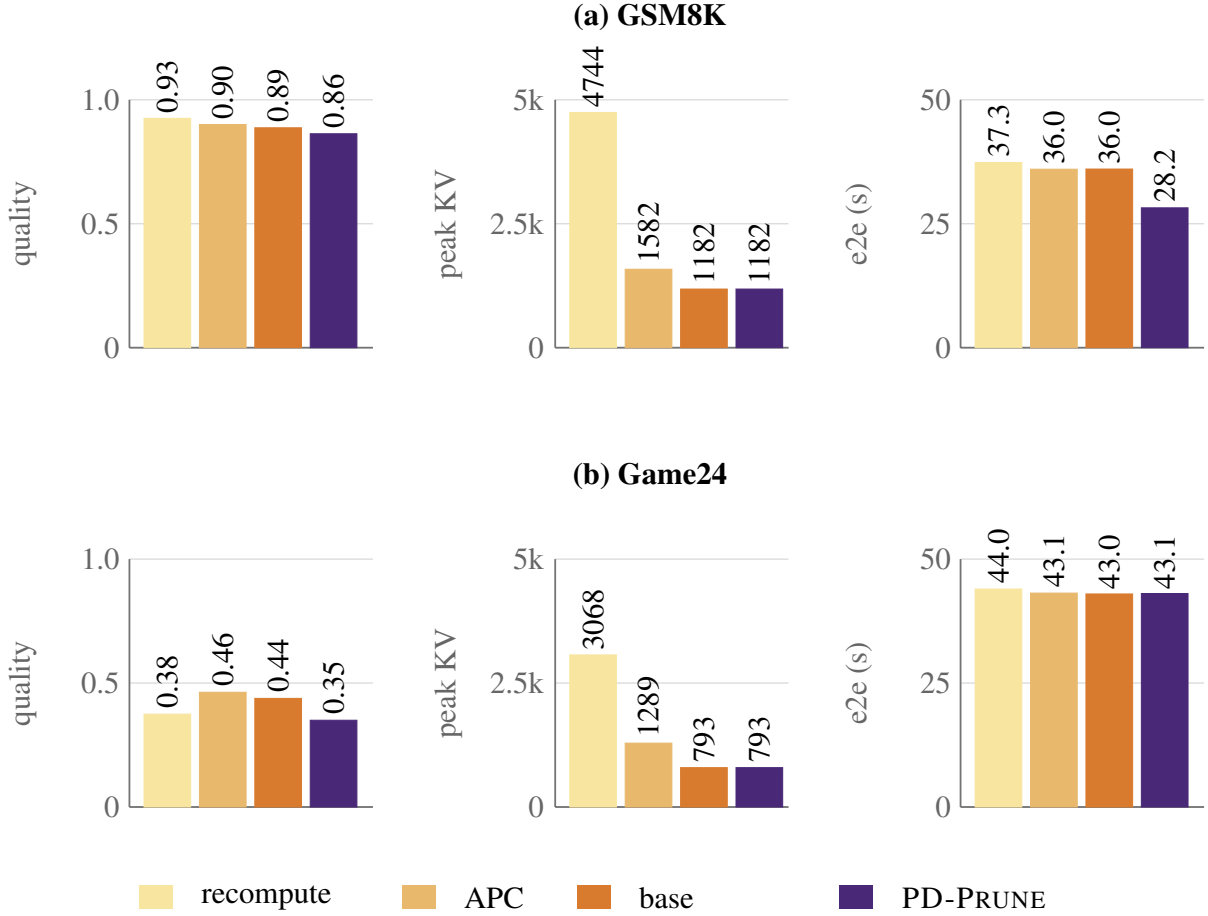


Figure 5: Four systems on Qwen3-8B ($n=80$, $tp=2$, $D=512$). Each panel has its own axis. Quality is higher-better; peak KV and end-to-end time are measured values, so a shorter bar is better. (a) GSM8K. (b) Game24. PD-PRUNE lowers peak KV below APC on both workloads. GSM8K quality stays with APC, and PD-PRUNE also lowers GSM8K end-to-end time.

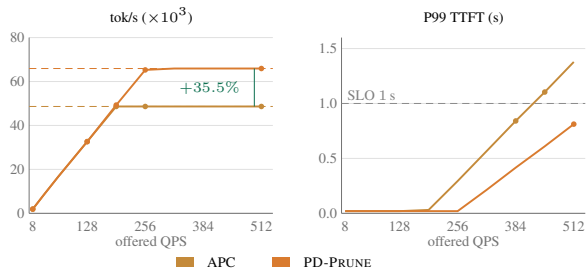


Figure 6: Offered-load sweep from the measured DeepSeek-R1 GSM8K peaks. Left: saturated tokens per second plateau at concurrency times the decode rate. Right: APC crosses the 1 s P99 line first; PD-PRUNE stays under it at a higher QPS.